# Sisteme de ecuații neliniare

$$\begin{cases} f_1(x_1, x_2, \dots x_n) = 0 \\ f_2(x_1, x_2, \dots x_n) = 0 \\ f_n(x_1, x_2, \dots x_n) = 0 \end{cases}$$

ex

$$\begin{cases} 3x_1 - \cos(x_2 x_3) - \frac{1}{2} = 0 \\ x_1^2 - 81(x_2 + 0{,}1)^2 + \sin x_3 + 1{,}06 = 0 \\ e^{-x_1 x_2} + 20x_3 + \frac{10\pi - 3}{3} = 0 \end{cases}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad F \overset{not}{=} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} \qquad F(x) = \begin{bmatrix} f_1(x) \\ f_2(x) \\ f_3(x) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

## a) Puncte fixe

$D \in \mathbb{R}^n \qquad x \in D$

$\exists \left| \frac{\partial f_i(x_1 \dots x_n)}{\partial x_j} \right| \leq K$ și $f_i(x_1^0 \dots x_n^0)$ continuă

$G(x) = x - F(x) \qquad x^{(0)}, G(x^0)$ — G - contracție

$$\left| \frac{\partial g_i(x)}{\partial x_j} \right| \leq \frac{K}{n}$$

ex

$D = \{ -1 \leq x_i < 1 \mid i = 1; 2; 3 \} \qquad x^{(0)} = [0{,}1 \quad 0{,}1 \quad -0{,}1]^T$

$$\begin{cases} x_1 = \frac{1}{3}\cos(x_2 x_3) + \frac{1}{6} \\ x_2 = \frac{1}{9}\sqrt{x_1^2 + \sin x_3 + 1{,}06} - 0{,}1 \\ x_3 = -\frac{1}{20}e^{-x_1 x_2} - \frac{10\pi - 3}{60} \end{cases} \qquad x = G(x) \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} g_1(x) \\ g_2(x) \\ g_3(x) \end{bmatrix}$$

$$|g_1(x_1,x_2,x_3)| \le \frac{1}{3} + \frac{1}{6} = \frac{1}{2} = 0{,}5$$

$$|g_2(x_1,x_2,x_3)| \le \frac{1}{9}\sqrt{1+\sin 1 + 1{,}06} - 0{,}1 < 0{,}09$$

$$|g_3(x_1,x_2,x_3)| \le \frac{1}{20}e + \frac{10\pi - 3}{60} < 0{,}61$$

$g_i(x) \in D$

$$\left|\frac{\partial g_1}{\partial x_1}\right| = 0 \qquad \left|\frac{\partial g_1}{\partial x_2}\right| = \left|\frac{1}{3}x_3 \sin(x_2 x_3)\right| \le \frac{1}{3}\sin 1 < 0{,}281$$

$$\left|\frac{\partial g_1}{\partial x_3}\right| < 0{,}281$$

$$\left|\frac{\partial g_2}{\partial x_2}\right| = 0 \qquad \left|\frac{\partial g_2}{\partial x_1}\right| < 0{,}238 \qquad \left|\frac{\partial g_2}{\partial x_3}\right| < 0{,}119$$

$$\left|\frac{\partial g_3}{\partial x_3}\right| = 0 \qquad \left|\frac{\partial g_3}{\partial x_1}\right| < 0{,}14 \qquad \left|\frac{\partial g_3}{\partial x_2}\right| < 0{,}14$$

$$\left|\frac{\partial g_i}{\partial x_j}\right| \le 0{,}281 \Rightarrow k = 3 \cdot 0{,}281 = 0{,}843 < 1$$

$$x^{(1)} = G(x^{(0)})$$

$$x^{(2)} = G(x^{(1)}) \quad \ldots\ldots \quad \|x^{(k+1)} - x^{(k)}\| < \varepsilon$$

pt $k = 5$ $\|x^{(5)} - x^{(4)}\|_\infty = 3.1 \times 10^{-7}$

## b) Newton-Raphson

$$\begin{cases} f_1(x_1, x_2, \dots x_n) = 0 \\ f_2(x_1, x_2, \dots x_n) = 0 \\ \\ f_n(x_1, x_2, \dots x_n) = 0 \end{cases} \qquad F(x) = \begin{bmatrix} f_1(x) \\ f_2(x) \\ \\ f_n(x) \end{bmatrix}$$

$$G(x) = X - J^{-1}(x) \cdot F(x)$$

$$J(x) = \begin{vmatrix} \frac{\partial f_1}{\partial x_1}(x) & \cdots & \frac{\partial f_1}{\partial x_n}(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1}(x) & \cdots & \frac{\partial f_n}{\partial x_n}(x) \end{vmatrix}$$

$$X^{(k+1)} = G(X^{(k)}) = X^{(k)} - J^{-1}(X^{(k)}) F(X^{(k)})$$

$\square \cdot \square$

$J \cdot \left| J^{-1}(X^{(k)}) F(X^{(k)}) = Y^{(k)} \right.$

$$F(X^{(k)}) = J(X^{(k)}) \cdot Y^{(k)}$$

$X^{(0)}, F, J_F$

pas k : $J(X^{(k)}) \cdot Y^{(k)} = F(X^{(k)})$ → S.E.L. (QR)

(±LU; GAUSS; J; GJ; SOR; G; GC)

$$X^{(k+1)} = X^{(k)} - Y^{(k)}$$

Rezolvarea unui sistem de ecuații neliniare presupune o aproximație inițială și rezolvarea la fiecare pas a unui sistem de ecuații liniare bazat pe Jacobiana sistemului neliniar și actualizarea iterației la pasul curent.

Obs

Jacobiana NU are stabilitate numerică